

# **CYBER SECURITY INTERNSHIP PROGRAM**

## **TASK 4: PASSWORD CRACKING**

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### **1. Objective**

The objective of this task was to understand password security by performing password hash cracking using John the Ripper. The task demonstrates how weak passwords can be recovered from hashed values using dictionary-based attacks and highlights the importance of strong password practices.

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### **2. Introduction**

Passwords are one of the most commonly used authentication mechanisms in computer systems. To protect user credentials, systems store passwords as hashes instead of plain text. However, weak passwords

can still be recovered using password auditing tools and wordlists.

John the Ripper is a widely used password auditing and recovery tool that helps security professionals evaluate password strength and identify weak credentials.

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### **3. Tools Used**

- Kali Linux
  - John the Ripper
  - Terminal
  - Custom Wordlist
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### **4. Installation of John the Ripper**

The installation was verified in Kali Linux using the following command:

```
john --version
```

The system confirmed that John the Ripper was already installed and updated to the latest available version.

### **Screenshot**

(Insert Screenshot: John the Ripper Installation)

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## 5. Creating a Password Hash

A sample password was selected and converted into an MD5 hash using the following command:

```
echo -n "password123" | md5sum
```

Generated Hash:

```
482c811da5d5b4bc6d497ffa98491e38
```

The generated hash was stored in a file named:

```
hashes.txt
```

### Screenshot

(Insert Screenshot: Password Hash Creation)

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## 6. Creating a Custom Wordlist

A custom password wordlist was created and stored in:

```
passwords.txt
```

Contents of the wordlist:

```
123456
```

```
admin
```

```
welcome
```

password123

qwerty

The wordlist contains commonly used passwords for testing password strength.

## **Screenshot**

(Insert Screenshot: Custom Wordlist Creation)

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## **7. Password Cracking Process**

John the Ripper was executed using the following command:

```
john --format=raw-md5 --wordlist=passwords.txt  
hashes.txt
```

The tool compared each password in the wordlist against the stored MD5 hash.

After processing the wordlist, John the Ripper successfully identified the correct password corresponding to the hash.

## **Screenshot**

(Insert Screenshot: Password Cracking Process)

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## 8. Results

The password hash was successfully cracked using a dictionary attack.

Recovered Password:

password123

The experiment demonstrated that weak and commonly used passwords can be recovered quickly when included in a wordlist.

### Screenshot

(Insert Screenshot: Cracked Password Result)

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## 9. Security Recommendations

1. Use strong and complex passwords.
  2. Avoid common dictionary words.
  3. Use multi-factor authentication (MFA).
  4. Implement password policies.
  5. Use secure password managers.
  6. Regularly update passwords.
  7. Store passwords using secure hashing algorithms.
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## **10. Conclusion**

This task provided practical experience in password auditing and hash cracking using John the Ripper. The exercise demonstrated how weak passwords can be recovered through dictionary attacks and emphasized the importance of creating strong, unique passwords. Understanding these techniques helps security professionals assess password strength and improve organizational security practices.

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### **Learning Outcome**

- Learned the concept of password hashing.
- Understood dictionary-based password attacks.
- Gained hands-on experience with John the Ripper.
- Understood the importance of strong password security practices.
- Developed practical skills in password auditing and security assessment.